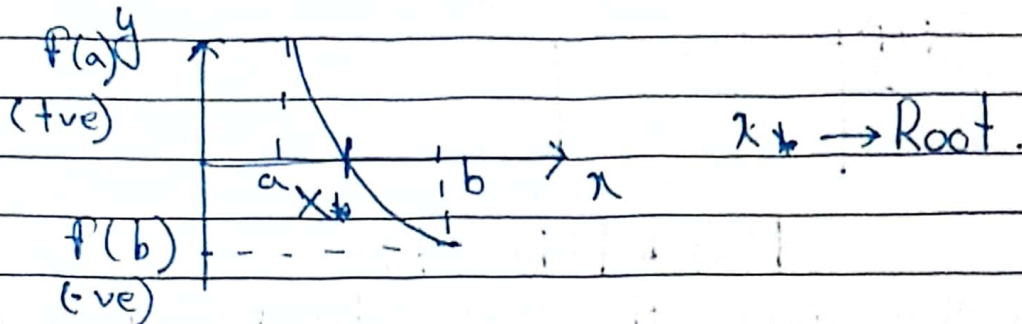


Bisection Method [Root finding Algorithm]

The solution of x is known as roots.



$$f(x) = x^2 - 2x$$

$$f(x) = 0$$

$$x(x-2) = 0$$

$$x = 0, x = 2$$

We cannot solve this ^{easily} always because sometimes the functions are infinitely large.

So, we use iterative method which tries to approximate a root. The solution is approximate not exact. This is Root Finding Algorithm.

In these type of problems, an interval will be given -

$$x \in [a, b]$$

The root somewhere lies between this interval.

The first thing to do is check the validity of the interval

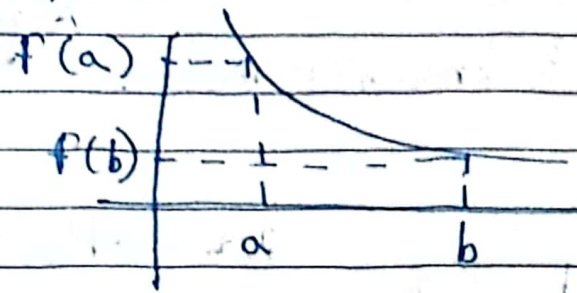
$$x \in [a, b] \rightarrow \text{Valid / Invalid}$$

$$f(a) * f(b) = -ve$$

$$+ve * -ve = -ve$$

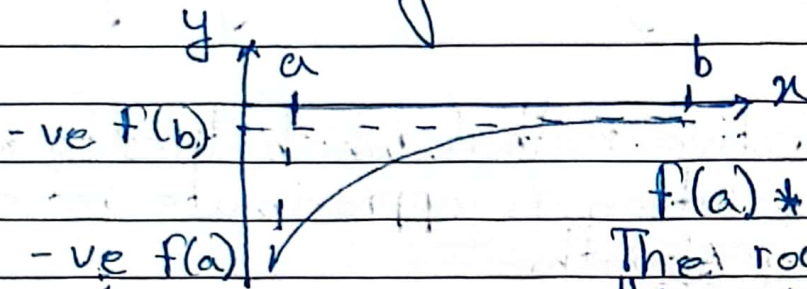
This is a valid range because $f(a) * f(b)$ is negative which indicates that the sign of y changed. So at some point the value of $y/f(x)$ became zero where we found the root.

For example :



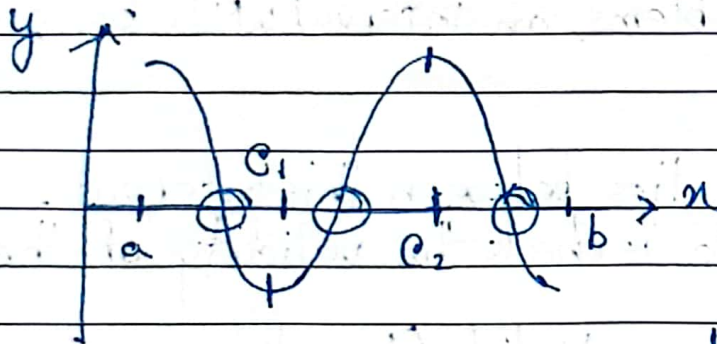
$$f(a) * f(b) = +ve$$

In this graph, the value of y never became zero so the graph never touched the y axis from where we could find the x root. This is not a valid range.



$$f(a) * f(b) = +ve$$

The root doesn't lie within this interval.

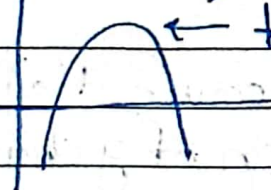


$[a, b]$

Multiple roots

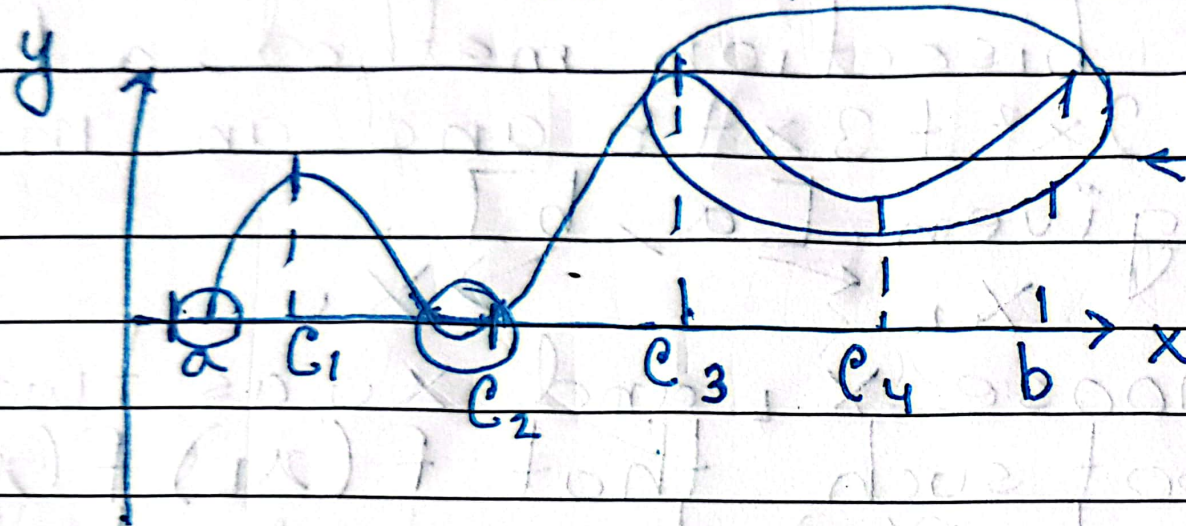
So we need to divide the given interval into sub-intervals

In order to do this, we need to label the turning points.



$$[a, b] = [a, c_1] \cup [c_1, c_2] \cup [c_2, b]$$

↳ sub-interval



Avoid this interval
because we can see
no root here.

$$[a, b] = [a, c_1] \cup [c_1, c_2] \cup [c_2, c_3]$$

Bisection Method [Root finding Algorithm]

When bisection method is going to be applied an equation will be given, $f(x) = 2x^2 + 3x + 5$ and an interval will be given, $[a, b]$.

Complete Algorithm for the Bisection Method

Step 1 : Choose x_l and x_u as two guesses for the roots such that $f(x_l)f(x_u) < 0$ in other words, $f(x)$ changes sign between x_l and x_u .

for the given range, $[a, b]$

$\swarrow \quad \nwarrow$
 $x_l \quad \quad x_u$
(x lower) (x upper)

$f(x_l)f(x_u) < 0$ checks the validity of the given interval $[a, b]$.

Step 2 : Estimate the root x_m of the equation as the mid-point between x_l and x_u , $x_m = \frac{x_l + x_u}{2}$
 $= \frac{a + b}{2}$

This is similar to binary search. The interval is divided and checks whether the interval lies in the left side or right side.

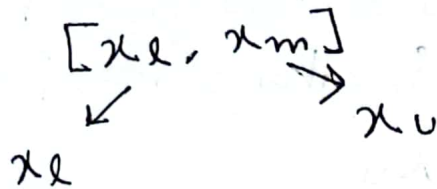
Step 3 : Now check the following

Case 1: If $f(x_l) \times f(x_m) < 0$ the new interval will be

$$x_l = x_l$$

$$x_u = x_m$$

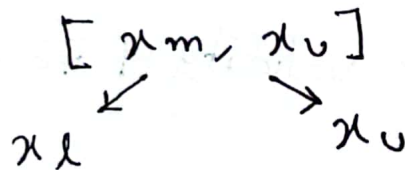
$$[x_l, \dots, x_m, \dots, x_u]$$



Case 2 : If $f(x_l) \times f(x_m) > 0$ the new interval will be

$$x_l = x_m$$

$$x_u = x_u$$



Case 3 : If $f(x_l) \times f(x_m) = 0$

so x_m is the root (This case does not generally occur but case 1 and case 2 is more frequently occurring)

Example : $f(x) = x^3 - 0.165x^2 + 3.993 \times 10^{-4}$

$[0, 0.11]$ ← given interval of the interval
 * * Firstly - we need to check the validity, but in question if it is not mentioned then no need to check it.

$$f(x_l) = 0^3 - 0.165(0)^2 + 3.993 \times 10^{-4} = 3.993 \times 10^{-4}$$

$$f(x_u) = (0.11)^3 - 0.165(0.11)^2 + 3.993 \times 10^{-4} \\ = -2.662 \times 10^{-4}$$

$f(x_l) f(x_u) < 0 \rightarrow$ The given interval is valid.

1st iteration:

$$x_m = \frac{0 + 0.11}{2} = 0.055$$

$$f(x_m) = (0.055)^3 - 0.165(0.055)^2 + 3.993 \times 10^{-4} \\ = 6.655 \times 10^{-5}$$

$$f(x_l) f(x_m) = (3.993 \times 10^{-4}) * (6.655 \times 10^{-5})$$

$$f(x_l) f(x_m) = +ve$$

$$f(x_l) f(x_m) > 0 \text{ [So Case 2 occurs here]}$$

$$x_l = x_m$$

$$x_l = 0.055$$

$$x_u = x_u$$

$$x_u = 0.11$$

New interval is $[0.055, 0.11]$

The root never lies in the interval $[0, 0.055]$ so we have avoided this interval.

2nd
iteration:

$$x_m = \frac{0.055 + 0.11}{2} = 0.0825$$

$$f(x_l) f(x_m) = f(0.055) f(0.0825)$$

$$f(x_l) f(x_m) = (6.655 \times 10^{-5}) (-1.622 \times 10^{-4})$$

$$f(x_l) f(x_m) = -ve$$

Case I $x_l = x_{\text{old}} = 0.055$
occurs here $x_0 = x_m = 0.0825$

New interval $[0.055, 0.0825]$

The iteration will continue till root is found.

★★ Question: How many iterations it will take to find the root of the function / Find Iteration - n?

[Note: The root we find will be with some error bound]
Interval will be given $x \in [1.5, 3]$ and
Machine Epsilon / will be given $= 1.1 \times 10^{-16}$
Scale invariant error

$$n \geq \frac{\log |b - a| - \log(\epsilon)}{\log 2} + 1$$

Note if accuracy is given then the formula will be:

$$n \geq \frac{\log |b - a| - \log(\epsilon)}{\log 2}$$

$$n \geq \frac{\log |3 - 1.5| - \log (1.1 \times 10^{-16})}{\log 2} - 1$$

$$n \geq 52.5983$$

$n \geq 53$ [Iteration can't be fractional]

If Accuracy was given 1.1×10^{-16} ,

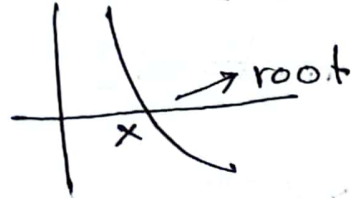
$$n \geq \frac{\log |3 - 1.5| - \log (1.1 \times 10^{-16})}{\log 2}$$

Drawback of Bisection Method:

It is comparatively slower.

Fixed Point Iteration [Root finding Algorithm]

So far we have seen the point at which $f(x) = 0$ which is the graph cuts the x -axis is the root.



$$f(x) = 0$$

In root fixed point iteration we will convert this $f(x)$ into $g(x)$ using algebraic manipulation.

$$g(x) = x \text{ [fixed point]}$$

↳ root of the actual $f(x)$

Example $f(x) = x^2 - 2x - 3$ $f(x) = 0$

1st way:

$$x^2 = 2x + 3 \quad [\text{Make } x \text{ the subject}]$$

$$x = \sqrt{2x + 3}$$

$$g_1(x) = \sqrt{2x + 3}$$

2nd way:

$$x^2 - 2x - 3 = 0$$

$$x^2 - x - x - 3 = 0$$

$$-x = x + 3 - x^2$$

$$x = x^2 - x - 3$$

$$g_2(x) = x^2 - x - 3$$

3rd way:

$$x^2 - 2x - 3 = 0$$

$$2x^2 - x^2 - 2x - 3 = 0$$

$$2x^2 - 2x = x^2 + 3$$

$$x(2x - 2) = x^2 + 3$$

$$x = \frac{x^2 + 3}{2x - 2}$$

$$g_3(x) = \frac{x^2 + 3}{2x - 2}$$

* Question might ask you to find the actual value

$$x^2 - x - 3 = 0$$

$$x^2 - 3x + x - 3 = 0$$

$$x(x - 3) + 1(x - 3) = 0$$

$$(x - 3)(x + 1) = 0$$

$$x = -1, x = 3$$

$$g_1(x) = \sqrt{2x + 3} \rightarrow \text{Convergent}$$

$$g_1(0) = 1.73$$

$$g_1(1.73) = 2.54$$

$$g_1(2.54) = 2.84$$

$$g_1(2.84) = 2.95$$

$$g_1(2.95) = 2.98$$

$$g_1(2.98) = 3.00$$

$$g_1(3.00) = 3.00$$

← root of $f(x)$

[Keep in mind significant figures and decimal places]

$$g_2(x) = x^2 - x - 3 \rightarrow \text{Divergent}$$

$$x_0 = 0$$

$$g_2(0) = -3$$

$$g_2(-3) = 9.0$$

$$g_2(9.0) = 69.0$$

$$g_2(69.0) = 4.69 \times 10^3$$

$$g_3(x) = \frac{x^2 + 3}{2x - 2} \quad x_0 = 0 \rightarrow \text{Convergent}$$

$$g_3(0) = -1.50$$

$$g_3(-1.50) = -1.05$$

$$g_3(-1.05) = -1.00$$

$$g_3(-1.00) = -1.00 \rightarrow \text{Fixed point}$$

Root of the
 $f(x)$

If we take $x_0 = 42$,

$$g(42) = 21.6$$

$$g(21.6) = 11.4$$

$$g(11.4) = 6.39$$

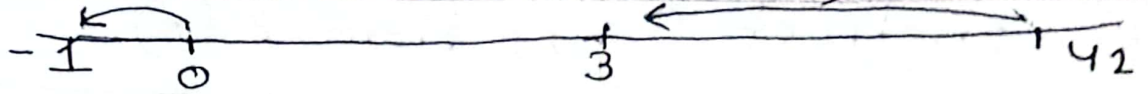
$$g(6.39) = 4.07$$

$$g(4.07) = 3.19$$

$$g(3.19) = 3.01$$

$$g(3.01) = 3.00$$

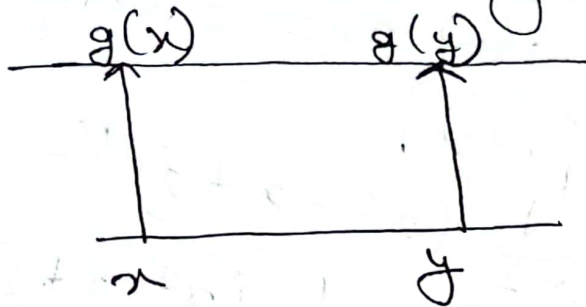
If we consider the number line,



$x_0 = 0$ [It is closer to -1 so converges to the nearest root]

$x_0 = 42$ [It is closer to 3 so converges to the nearest root]

Contraction Mapping Theorem:



$$\lambda = \left| \frac{g(y) - g(x)}{y - x} \right|$$

$$\lambda = |g'(x)| \text{ at the roots}$$

$\lambda = 0$ [Super linear Convergent]

$0 < \lambda < 1$ [Linear Convergent]

$\lambda \geq 1$ [Divergent]

[It indicates whether the function is convergent or divergent.]

$$f(x) = x^3 - 2x^2 - x + 2$$

@ Find the roots of the solution.

$$x^2(x-2) - 1(x-2) = 0$$

$$(x^2 - 1)(x-2) = 0$$

$$x = 1, -1, 2$$

(b) Construct $3g(x)$ from $f(x)$

$$g_1(x) = \sqrt{1/2(x^3 - x + 2)}$$

$$g_2(x) = x^3 - 2x^2 + 2$$

$$g_3(x) = \frac{-2}{x^2 - 2x - 1}$$

(c) Determine which $g(x)$ are convergent and which $g(x)$ are divergent.

$$\lambda = |g'_1(x)|$$

$$g_1(x) = \frac{1}{\sqrt{2}} (x^3 - x + 2)^{1/2}$$

$$g'_1(x) = \frac{1}{\sqrt{2}} \cdot \frac{1}{2} (x^3 - x + 2)^{-1/2} (3x^2 - 1)$$

$$g'_1(x) = \frac{(3x^2 - 1)}{2\sqrt{2} (x^3 - x + 2)^{1/2}}$$

$$\lambda = |g'_1(x)| = \left| \frac{3x^2 - 1}{2\sqrt{2} (x^3 - x + 2)^{1/2}} \right|$$

$$x = 1 \rightarrow 0.5 \text{ (linear convergence)}$$

$$x = -1 \rightarrow 0.6 \text{ (linear convergence)}$$

$$x = 2 \rightarrow 3.75 \text{ (divergent)}$$

$$\lambda = |g_2'(x)| = |3x^2 - 4x| = \text{[check } \lambda \text{ values for } x = -1, 1, 2]$$

$$\begin{array}{l} x = -1 \rightarrow 7 \geq 1 \text{ [Divergent]} \\ x = 1 \rightarrow 1 \geq 1 \text{ [Divergent]} \\ x = 2 \rightarrow 4 \geq 1 \text{ [Divergent]} \end{array}$$

$$\lambda = |g_3'(x)| = \left| -\frac{2}{x^2 - 2x - 1} \right|$$

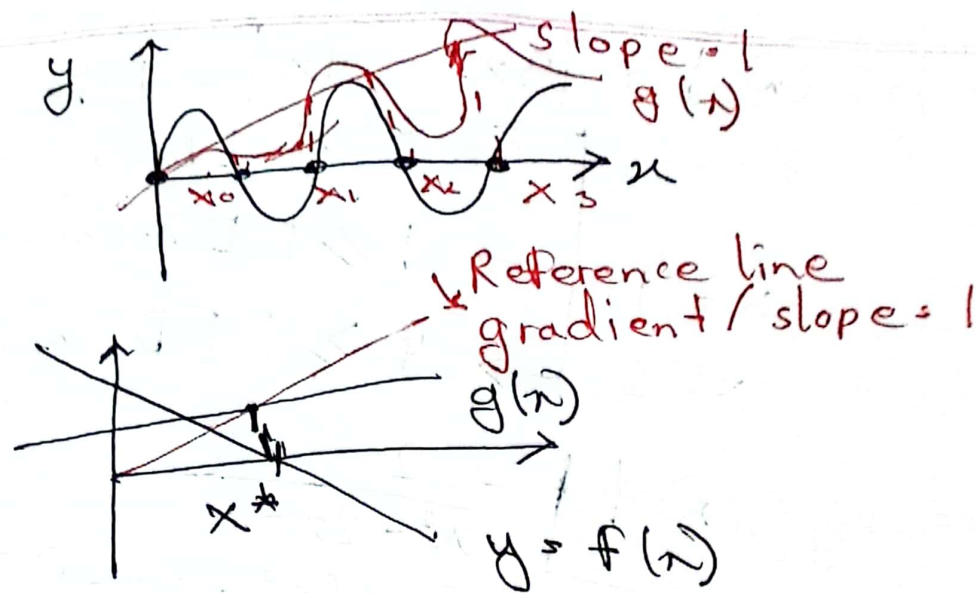
$$= -2 \times (x^2 - 2x - 1)^{-1}$$

$$= -2 \times -1 (x^2 - 2x - 1)^{-2} (2x - 2)$$

$$= \frac{2(2x - 2)}{(x^2 - 2x - 1)^2}$$

$$\lambda = |g_3'(x)| = \left| \frac{2(2x - 2)}{(x^2 - 2x - 1)^2} \right|$$

$$\begin{array}{l} x = -1 \rightarrow 2 \geq 1 \text{ [Divergent]} \\ x = 1 \rightarrow 0 \text{ [Super linear convergent]} \\ x = 2 \rightarrow 4 \geq 1 \text{ [Divergent]} \end{array}$$



The points at which the $g(x)$ graph cuts the slope = 1 line / reference line we get the actual roots for the function at those points

Chapter 04

No. of iterations : [Derivation]

$$f(x) = \frac{1}{x} - 0.5 \quad [1.5, 3] = \text{Interval}$$

$$a_0 = 1.5, \quad b_0 = 3, \quad m_0 = 2.25$$

$$f(a_0) * f(m_0) = [0.166 * 0.055] < 0$$

root lies between a_0 and m_0

$$a_1 = a_0 = 1.5, \quad b_1 = m_0 = 2.25$$

$$m_1 = \frac{a_1 + b_1}{2} = 1.875 \quad f(a_1) * f(m_1) = [0.166 * 0.033] > 0$$

root lies between m_1 and b_1

Now, using the values from the maths,

$$|b_0 - a_0| = |3 - 1.5| = 1.5$$

$$|b_1 - a_1| = |2.25 - 1.5| = 0.75$$

We can find a pattern here,

$$|b_1 - a_1| = \frac{|b_0 - a_0|}{2}$$

$$|b_2 - a_2| = \frac{|b_1 - a_1|}{2}$$

Similarly we can write, $|b_2 - a_2| = \frac{|b_0 - a_0|}{2 * 2}$

$$|b_k - a_k| = \frac{|b_0 - a_0|}{2^k} \quad \text{Eqn (1)}$$

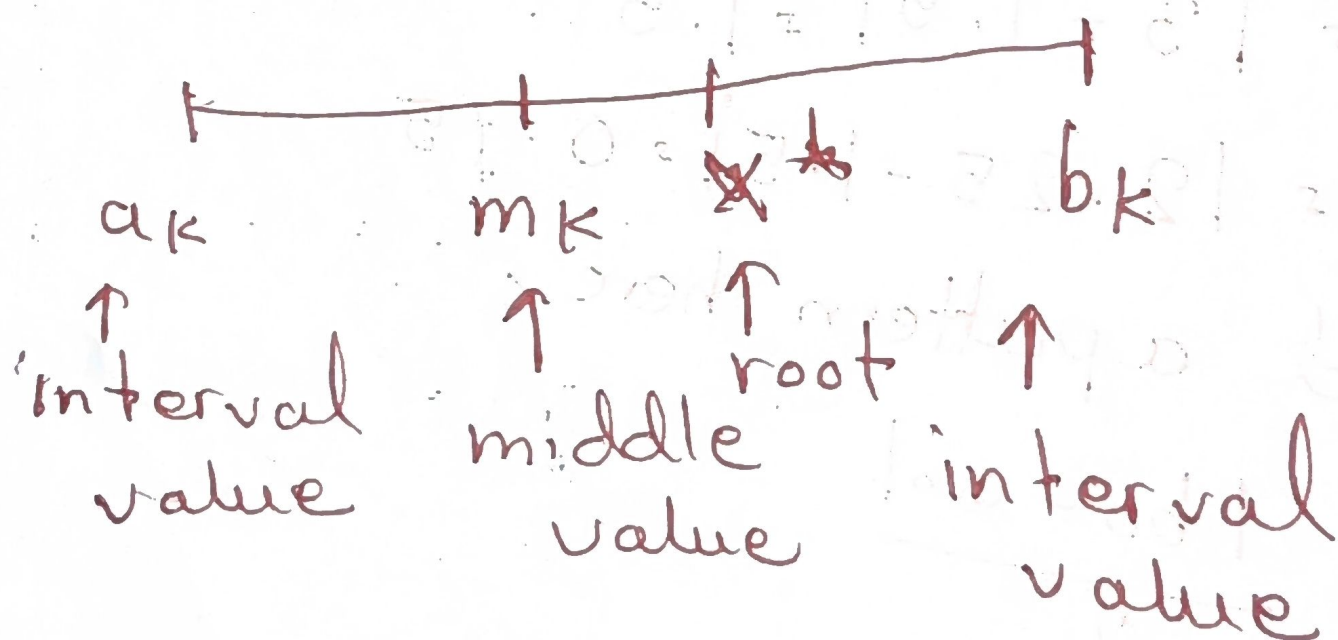
We know that x^* is the actual root but in bisection method we find the approximate root which

is m_k .

So if we want to find the error

then $|m_k - x^*| = \text{Error}$

If we use a line to plot out the values,



$$|m_k - x^*| \leq \frac{|b_k - a_k|}{2}$$

$$|m_k - x^*| \leq \frac{\left| \frac{b_0 - a_0}{2^k} \right|}{2}$$

$$\begin{array}{ccc} |m_k - x^*| & \leq & \frac{|b_0 - a_0|}{|2^{k+1}|} \rightarrow \text{Maximum possible error, we can consider it as } \delta_{\max}/\epsilon \\ \uparrow & & \\ \text{Actual Error} & & \end{array}$$

$$|m_k - x^*| \leq \delta \cdot \frac{|b_0 - a_0|}{2^{k+1}}$$

$$\left| \frac{b_0 - a_0}{2^{k+1}} \right| \leq \delta$$

Now our target is to find the value of k ,
Taking log on both sides,

$$|b_0 - a_0| \leq \delta \cdot 2^{k+1}$$

$$\frac{|b_0 - a_0|}{\delta} \leq 2^{k+1}$$

$$\log\left(\frac{|b_0 - a_0|}{\delta}\right) \leq \log(2^{k+1}) \Rightarrow \log\left(\frac{|b_0 - a_0|}{\delta}\right) \leq (k+1)\log 2$$

$$\frac{\log\left(\frac{|b_0 - a_0|}{\delta}\right)}{\log 2} - 1 \leq k$$

In our notes since we have used n as notation for no. of iterations we can write,

$$n \geq \frac{\log |b-a| - \log \delta}{\log 2} - 1$$

For accuracy -

$$n \geq \frac{\log |b-a| - \log \delta}{\log 2} + 1$$